

Documentation:

Student Performance Analytics System

1. Objectives

- **Automated Data Processing:** Load student academic records and compute total marks, average marks, letter grades, and pass/fail statuses using NumPy and Pandas.
- **Performance Analysis:** Evaluate overall class performance, subject-wise averages, and individual achievements.
- **Reporting:** Provide a clean, structured summary of student metrics, including top performers, subject leaders, and pass/fail distribution.

2. Implementation

The project is split into two core modules:

- **functions.py:** Contains helper functions for data manipulation and analytics:
 - **calculate_totals_and_averages(df):** Computes row-wise total and average marks across four subjects (Maths, Science, English, Computer_Science).
 - **assign_grade(avg):** Assigns letter grades (A, B, C, D, F) based on average marks thresholds (85+, 70+, 55+, 40+, and below 40 respectively).
 - **check_pass_fail(avg, attendance):** Determines a student's status (Pass or Fail) requiring a minimum average of 40 and attendance of 65%.
 - **process_student_data(df):** Orchestrates the transformation pipeline.
 - **get_subject_averages(df) & get_top_performers(df, n):** Aggregate performance metrics and query top students.
- **main.py:** The execution entry point that loads **students.csv**, invokes processing functions, and outputs formatted console reports.

3. Final Outcome

Executing the analytics script on the 100-student dataset yielded the following results:

- **Dataset Overview:** Successfully loaded 100 records from **students.csv**.
- **Overall Class Performance:**
 - **Overall Class Average Marks:** 77.11
 - **Total Students Passed:** 99
 - **Total Students Failed:** 1
- **Subject-wise Average Performance:**
 - **Computer Science:** 80.58 (Best Performing Subject)
 - **Science:** 76.77
 - **Maths:** 75.74
 - **English:** 75.35
- **Top Achievers:**
 - **Highest Average Student:** Varun Kapoor (Student ID: 191) with an average of 95.00.